Course Code	Course Title	L	T	P	C
MACSE513	**Computer Networks**	**3**	**0**	**2**	**4**
Pre-requisite	Nil	**Syllabus Version**			
		1			

Course Objectives

1. To impart knowledge of various network models, their layered architecture, and associated protocols.
2. To equip with the foundational knowledge of routing algorithms and their applications in networking.
3. To facilitate the comprehension of the fundamentals, architecture, and characteristics of wireless and mobile networks.

Course Outcomes

1. Explore the Internet model and various network performance metrics.
2. Interpret the application layer protocols, services and evaluate the requirements for reliable services and implications of congestion at the transport layer services.
3. Analyze various functionalities required in the control and data plane at network layer services.
4. Explain the functions of data link layer including error control and access control.
5. Evaluate the characteristics of wireless and mobile networks, including their security protocols and standards.

Module:1	**Computer Networks and Internet**	**9 hours**

Internet: A Nuts-and-Bolts Description - Network Protocols - The Network Edge: Access Networks and Physical Media - The Network Core: Packet Switching, Circuit Switching - Network of Networks - Delay, Loss and Throughput in Packet-Switched Networks – Protocol Layers and Their Service Models(TCP/IP)- Principles of Network Applications: Architectures

Module:2	**Application Layer and Transport Layer**	**10 hours**

Application Layer: The Web and HTTP - Electronic Mail in the Internet, FTP - DNS—The Internet's Directory Service - Peer-to-Peer File Distribution - Transport Layer: Relationship Between Transport and Network Layers - Overview of the Transport Layer in the Internet - Multiplexing and Demultiplexing - Connectionless Transport: UDP – Reliable Data Transfer: Go-Back-N (GBN) and Selective Repeat (SR) - Connection-Oriented Transport: TCP, Flow Control and Congestion Control - Socket Programming: Creating Network Applications: TCP/UDP socket programming

Module:3	**Network Layer**	**10 hours**

Network Layer – Router - The Internet Protocol (IP): IPv4, Addressing and IPv6 – Generalized Forwarding and SDN - Control Plane: Per-router control and logically centralized control - Routing Algorithms - Link-State (LS) Routing Algorithm,

Distance-Vector (DV) Routing Algorithm, Intra-AS Routing in the Internet: OSPF and Routing Among the ISPs: BGP - SDN Control Plane		
Module:4	**Link Layer and LANs**	**8 hours**
Overview of Link Layer Services - Error-Detection and -Correction Techniques: Checksum and CRC - Multiple Access Links and Protocols: Channel Partitioning Protocols and Random-Access Protocols - Switched Local Area Networks: Link-Layer Addressing and ARP - Virtual Local Area Networks		
Module:5	**Wireless and Mobile Networks Security**	**6 hours**
Elements of a wireless network - Wireless Links and Network Characteristics - WiFi: 802.11 Wireless LANs - Mobility Management: Principles - Wireless and Mobility: Impact on Higher Layer Protocol- Security in Computer Network- Message Integrity and Digital Signatures - Network-Layer Security: IPsec and Virtual Private Networks		
Module:6	**Contemporary Issues**	**2 hours**
	Total Lecture Hours:	**45 hours**

Text Book(s)

James F. Kurose, Keith W. Ross, "**Computer Networking: A Top-Down Approach**", Pearson, United Kingdom., 8th Edition, **2022**

Reference Books

Andrew S. Tanenbaum, "**Computer Networks**", Pearson, Singapore, 6th Edition, **2022**
Larry Peterson and Bruce Davie, "**Computer Networks: A Systems Approach**", Morgan Kaufmann, United States of America., 6th Edition, **2021**
Larry Peterson and Bruce Davie, "**Computer Networks: A Systems Approach**", Morgan Kaufmann, United States of America, 6th Edition, **2021**

Indicative Experiments

1. Hardware Demo (Demo session of all networking hardware and Functionalities) OS Commands (Network configuration commands): □ □ □ □□Identify and understand the functionalities of networking hardware (e.g., routers, switches, hubs, cables). □ Execute network configuration commands (e.g., ifconfig, ipconfig, ping, tracert, netstat) on operating systems like Linux/Windows to configure and troubleshoot network settings.	2 hours

□ □□Document the purpose and output of at least ten commands and describe the role of each hardware component.	
2. Network Packet Analysis using Wireshark: □ □ □ □ □ □ □ □ □ □**Packet Capture**: Capture live network traffic from a specified interface (e.g., Ethernet, Wi- Fi) for at least 5 minutes. □ □ □ □ □**Starting Wireshark**: Configure Wireshark with appropriate capture filters to focus on specific protocols (e.g., HTTP, TCP). □ □ □ □ □□**Viewing Captured Traffic**: Analyze captured packets, identifying key fields like source/destination IP, protocol, and payload. □ □ □ □ □**Analysis, Statistics, and Filters**: Generate statistical reports (e.g., protocol hierarchy, conversations) and apply display filters to isolate specific traffic (e.g., tcp.port == 80).	3 hours
3. Socket programming (TCP and UDP), Multi-client chatting: □ □ □ □□Develop a client-server application using socket programming in C language. Implement a TCP/UDP based server that handles multiple clients simultaneously for a chat application. □ □ □ □ □Implement a TCP/UDP based client-server application for simple message exchange. □ □ □ □ □Test the applications by running multiple clients and verifying message delivery.	4 hours
4. IP addressing, Classless addressing: □ □ □ □ □Configure IP addresses on a simulated network using tools like Cisco Packet Tracer. □ □ □ □ □Assign IP addresses to devices using both classful and classless (CIDR) addressing schemes. □ □ □ □ □Calculate subnets for a given network (e.g., divide a /24 network into subnets). □ □ □ □ □Verify connectivity using ping and document the addressing scheme. □□□□□□ Implement a simple client-server program where the server listens on a configured IP address and port. The client sends a message containing classless IP addresses to the server. The server	3 hours

calculates the first address, last address, and other related information. Test the program to ensure successful communication.	
5. Performance evaluation of routing protocols using simulation tools: □ □ □ □□Use the NS-3 simulator to evaluate the performance of routing protocols (e.g., RIPng, OSPFv3). □ □ □Write a C++ program in NS-3 to set up a network topology with at least five routers, configuring IPv6 addresses and enabling routing protocols (e.g., RIPng and OSPFv3). □ □ □ □Simulate traffic using NS-3's traffic generation tools (e.g., OnOffApplication) and measure performance metrics such as convergence time (using routing table updates), packet loss, and throughput (using FlowMonitor). □ □ □ □ □□Analyze the routing tables generated by each protocol to determine the shortest path between nodes. Compare the performance of at least two protocols (e.g., RIPng vs. OSPFv3) based on the collected metrics.	3 hours
6. SDN Applications and Use Cases Explore Software-Defined Networking (SDN) using a tool like Mininet. □□□□□□□ Create a simple SDN topology with a controller (e.g., OpenDaylight, Ryu). •□□□□□□□□ Implement a use case, such as traffic engineering or load balancing. •□□□□□□□□ Test the setup by generating traffic and analyzing controller behavior.	2 hours
7. Error detection and correction mechanisms, Flow control mechanisms: □ □ □ □□Implement and analyze error detection/correction and flow control mechanisms. □ □ □ □ □Write a program to simulate error detection (e.g., CRC, checksum) and correction (e.g., Hamming code) on sample data. □ □ □ □ □Implement a flow control mechanism (e.g., sliding window protocol) in a simulated environment. □ □ □ □ □Test the implementations with different error rates and window sizes.	4 hours

8. Wired and Wireless Network Simulation Using Cisco Packet Tracer: □□□□□□Use Cisco Packet Tracer to design and simulate a network. □□□□□Create a topology with both wired (e.g., Ethernet) and wireless (e.g., Wi-Fi) components, including at least two routers, two switches, and five end devices. □□□□□Configure network settings (e.g., IP addresses, VLANs, SSIDs). Test connectivity and simulate data transfer.	3 hours
9. Security in Network - Use cases: □□□□□Explore network security use cases. □□□□□Set up a network in a simulation tool (e.g., GNS3, Packet Tracer) and implement at least two security mechanisms (e.g., ACLs, VPN, firewall rules). □□□□□Simulate an attack (e.g., unauthorized access attempt) and demonstrate how the security measures prevent it. □□□□□Document the setup and results.	3 hours
10. Network Security with IPsec and Message Integrity: □□□□□Use GNS3 to configure a secure network and implement message integrity checks. □□□□□Set up a network topology with two routers and two end devices, configuring an IPsec VPN tunnel between the routers to secure communication using ESP (Encapsulating Security Payload) in tunnel mode. □□□□□Write a C program to implement message integrity using HMAC-SHA256 for a simple client-server application running on the end devices. The program will compute and verify digital signatures for messages exchanged over the VPN. □□□ Simulate data transfer through the VPN and verify that the IPsec tunnel encrypts traffic (using Wireshark to inspect packets). Test the HMAC implementation by introducing tampered messages and checking detection. Submit the GNS3 configuration file, the C source code for the HMAC implementation, a report detailing the IPsec setup, packet capture analysis, and results of the message integrity tests.	3 hours

Submit the GNS3 configuration file, the C source code for the HMAC implementation, a report detailing the IPsec setup, packet capture analysis, and results of the message integrity tests.	
Total Laboratory Hours:	**30 hours**

Text Book(s)
James F. Kurose, Keith W. Ross, " **Computer Networking: A Top-Down Approach**", Pearson, United Kingdom, 8th Edition, **2022**
Reference Books
Andrew S. Tanenbaum, "**Computer Networks**", Pearson, Singapore, 6th Edition, **2022**
Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment, Oral Examination, Seminar, Group Discussion

Recommended by Board of Studies :	23-05-2025
Approved by Academic Council : No. 78	12-06-2025